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Chaos 36, 033117 (2026)

<https://doi.org/10.1063/5.0314081>



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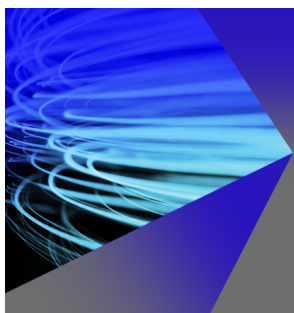
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Cite as: Chaos 36, 033117 (2026); doi: 10.1063/5.0314081

Submitted: 26 November 2025 · Accepted: 20 February 2026 ·

Published Online: 6 March 2026



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ABSTRACT

Reservoir computing (RC) is a powerful framework for predicting the temporal evolution of variables of nonlinear dynamical systems, yet the role of reservoir topology—particularly symmetry in connectivity and weights—remains not adequately understood. This work investigates how the structure of the reservoir network influences the performance of RC on time series prediction tasks derived from four dynamical systems of increasing complexity: the Mackey–Glass system with delayed-feedback, two nonlinear models of two-dimensional thermal convection flows, and a three-dimensional shear flow model exhibiting transition to turbulence. Using five reservoir topologies in which connectivity patterns and edge weights are controlled independently, we evaluate both direct- and cross-prediction tasks. The results show that symmetric reservoir networks improve prediction accuracy for the convection-based systems when the input dimension is smaller than the number of degrees of freedom. In contrast, the shear-flow model displays almost no sensitivity to topological symmetry due to its strongly chaotic high-dimensional dynamics. These findings reveal how structural properties of reservoir networks affect their ability to learn complex dynamics and provide guidance for designing more effective RC architectures.

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Predicting the evolution of nonlinear dynamical systems remains a central challenge across science and engineering. This study shows that the symmetry of the network topology of a reservoir computer—defined by both its node connections and connection weights—has a decisive impact on prediction accuracy. When the input dimension is smaller than the number of degrees of freedom of the target system, symmetric reservoir topologies yield superior performance due to enhanced cross prediction between variables. In contrast, asymmetric reservoirs perform better only when provided with full information from all degrees of freedom. These results demonstrate how the structure of reservoir networks fundamentally shapes the predictive capabilities of the

model and offers new pathways for designing more effective reservoir computing architectures.

I. INTRODUCTION

Reservoir computing (RC) has become a promising brain-inspired paradigm for recurrent data processing among machine-learning and neural-network architectures.^{1–4} Its inherent recurrence generates short-term memory and enables the prediction of complex temporal evolutions, allowing RC when implemented as an Echo State Network (ESN) to serve as a surrogate model for

nonlinear dynamical processes.⁵ RC has been applied widely across engineering and the natural sciences,^{6–8} including neuroscience, complex systems, machine learning, and fluid dynamics.^{3,9–14} Compared with deep recurrent neural networks, RC can be computationally cheaper while maintaining high prediction accuracy.^{15,16} A wide range of physical reservoir realizations have been explored, including water-bucket systems,⁵ Ising-spin networks,¹⁷ and parametric quantum circuits.¹⁸ As a result, RC remains an active research area, particularly with respect to physical implementations.^{5,19,20}

Despite this progress, key structural questions remain open for RC models in the form of an ESN. For example, the size of the reservoir is crucial: it must be large enough to capture the complexity of the target system while avoiding unnecessary computational cost,⁴ raising the question of what the minimal reservoir size is for a given problem. In particular, it remains poorly understood how strongly the topology and connectivity of the reservoir influence performance.²¹ Existing studies investigate various structural aspects, including directionality, sparsity, average path length, clustering, and degree distributions.^{4,21,22} Reservoirs with ring, lattice, random, small-world,^{23–26} scale-free,²⁷ and modular structures²⁸ have been explored. Notably, some studies report that for small reservoirs, uncoupled nodes may enhance long-term prediction and chaotic-attractor reconstruction.^{29–32} Recent results also show that random connectivity can outperform structured topologies, such as small-world networks when predicting Mackey–Glass time series.³³ Overall, however, the relationship between reservoir topology and performance remains unresolved, and no general principle yet links network structure to predictive capability.^{34,35}

Symmetries in the RC dynamics are considered adverse to train the RC for the desired task. For instance, a symmetry may result in mirror-attractors in the state space of RC and needs to be addressed.^{36,37} These symmetries may be broken in all building blocks of the RC model: (1) by shifting the input of a task (input-bias), (2) by the asymmetric structure of the network, and (3) by the output layer with output-bias or an addition of a quadratic read-out layer.^{36,38,39} The performance of an RC model was shown to be improved if symmetry of the input data is respected by the RC model, either in the input or in the output layer.³⁸

In fluid flows, particularly turbulent fluid flows that we are interested in, symmetries of the data, such as an inversion symmetry, are missing or established in a statistical sense only. Thus, tailoring input and output layers is not physically relevant. However in order to break inversion symmetry, the nonlinear activation function, such as $\tanh(\cdot)$ used here, requires a bias in its argument. In this work, however, we focus on the symmetry of the reservoir network structure with respect to connections and weights and will break these symmetries separately. This is different from a recent study by Carroll and Pecora³⁹ who considered a simple network graph with many symmetries. These symmetries manifest for permutations of the nodes (along with their edges) that leave the structure of the network preserved. In particular, when the reservoir network has only one graph-theoretic symmetry, with respect to flipping the sign of the elements in the adjacency matrix, a higher covariance rank of the reservoir state matrix follows and testing errors become comparable to those of a completely random reservoir network.

In the present work, we investigate the structure of reservoir networks for predicting the dynamics of four nonlinear dynamical

systems; three of them are derived as low-dimensional Galerkin models from fluid dynamical systems, such as buoyancy- and shear-driven fluid flows. The first two systems stand for truncations of thermal convection processes in a two-dimensional fluid layer, the Lorenz 63 (L63) model and its eight-dimensional extension (L8) that includes energy and vorticity conservation.^{40,41} The third system is a time-dependent three-dimensional shear flow (SF), which transitions from the laminar state to turbulence.⁴² System No. 4 is a Mackey–Glass (MG) equation.⁴³ While systems MG and L63 are frequently used and very popular benchmark cases, the higher-dimensional dynamical systems L8 and SF are significantly less investigated in the context of RC. However, they turn out to be appropriate candidates to develop RC-based surrogate models for more complex turbulent buoyancy-driven and wall-bounded flows.

We compare the performance of all four dynamical systems in a dynamics prediction task for five different topologies of the reservoir network in which network connectivity and weights along the network edges are chosen separately, similar to Ref. 33. This sums up to 20 supervised learning tasks with individual hyperparameter optimization. We find that the dynamics prediction, which comprises of direct- and cross-prediction tasks, is improved if reservoir network is symmetric for the first two systems. However, dynamics prediction of the third system (SF) is insensitive to the symmetry of the topology owing to the chaotic dynamics in higher-dimensional space.

The outline of the paper is as follows. Section II discusses the reservoir computing model approach and presents the different network topologies. In Sec. III, we present the four dynamical systems of increasing complexity that we study here. In Sec. IV, we summarize their performance in predicting the dynamics with partial state information provided to the reservoir. This includes a comparison of the five different topologies for the different dynamical systems. We summarize our analysis and give an outlook in Sec. V. Further details on the two higher-dimensional models L8 and SF are found in the Appendices A–D.

II. RESERVOIR NETWORK TOPOLOGIES

In the reservoir computing model, a target output $y(t)$ is predicted using its linear relationship with the reservoir state $r(t)$ that is driven by the input $u(t)$. The reservoir state $r(t)$ is a vector representing the state of N neurons and evolves in discrete time steps Δt , i.e., $t = k\Delta t$ (see IV A), as follows:

$$r(k+1) = (1 - \varepsilon)r(k) + \varepsilon \tanh[Wr(k) + W^{\text{in}}u(k+1) + b], \quad (1)$$

where W^{in} is an initially fixed random matrix to map an N_{in} -dimensional input vector $u(k+1)$ to the reservoir, W is the reservoir coupling matrix, and ε is the leaking rate. The state evolves with a bias represented by vector b in the nonlinear term, such that each element of b is sampled from a uniform random distribution $\mathcal{U}([-0.5, 0.5])$. This bias vector remains unchanged for subsequent training and testing phases. The predicted output y^p is given by the linear relationship

$$y^p(k+1) = W^{\text{out}}r(k+1). \quad (2)$$

The matrix W^{out} is the optimized output matrix. Only this matrix is trained with respect to the target output $y(k+1)$ using the ridge

regression scheme with a Tikhonov regularization parameter γ . The elements of the matrix W^{in} are randomly chosen from a uniform distribution $\mathcal{U}([-0.5, 0.5])$. The reservoir state is evolved for the first t_0 time steps during which it depends on the initial reservoir state (known as wash-out).^{1,15} The states $\mathbf{r}(k < k_0)$ are discarded as transients, which depend on the spectral radius of W , the leaking rate and the prediction task. The initial transient of k_0 steps depends on reservoir type; we take a sufficient number of $k_0 = 500$ steps for all reservoir types and tasks. The states $\mathbf{r}(k_0 < k < k_t)$ are collected and used to train the RC model output layer with matrix W^{out} such that $\mathbf{y}^p(k) = W^{\text{out}}\mathbf{r}(k)$. The output layer is further optimized for three out of the five hyperparameters—spectral radius, leaking rate, and Tikhonov regularization parameter. The optimized output matrix is employed to test the prediction for $k_t < k < k_p$ in a test phase of $k_p - k_t$ time steps. For all tasks in this work, $k_t = 2500$

and $k_p = 4500$ time steps. This means that we have 2000 training and testing steps. The matrix W is sparse representing a small fixed density of active nodes in all networks in this study.

We define a dynamics prediction task (DPT) as the prediction of the complete system state $\mathbf{y}(k) = [y_1, y_2, \dots, y_M]^T \in \mathbb{R}^M$ of the target dynamical system at the next time instant $(k+1)\Delta t$. In other words, a DPT consists of M tasks to predict M time series of the components of $\mathbf{y}(k)$ in successive one-step predictions. The RC model can be trained using either (i) all M time series $\mathbf{u} = [u_1, u_2, \dots, u_M]^T \in \mathbb{R}^M$ or (ii) $m < M$ time series $\mathbf{u} = [u_1, u_2, \dots, u_m]^T \in \mathbb{R}^m$ as the input to predict the next full system state, including the unseen data of remaining dimensions $M - m$. This is also known as the open-loop scenario of reservoir computing since the ground truth input is fed at each step.⁴⁴ In the case of (i), the prediction is expected to be easier, since the complete

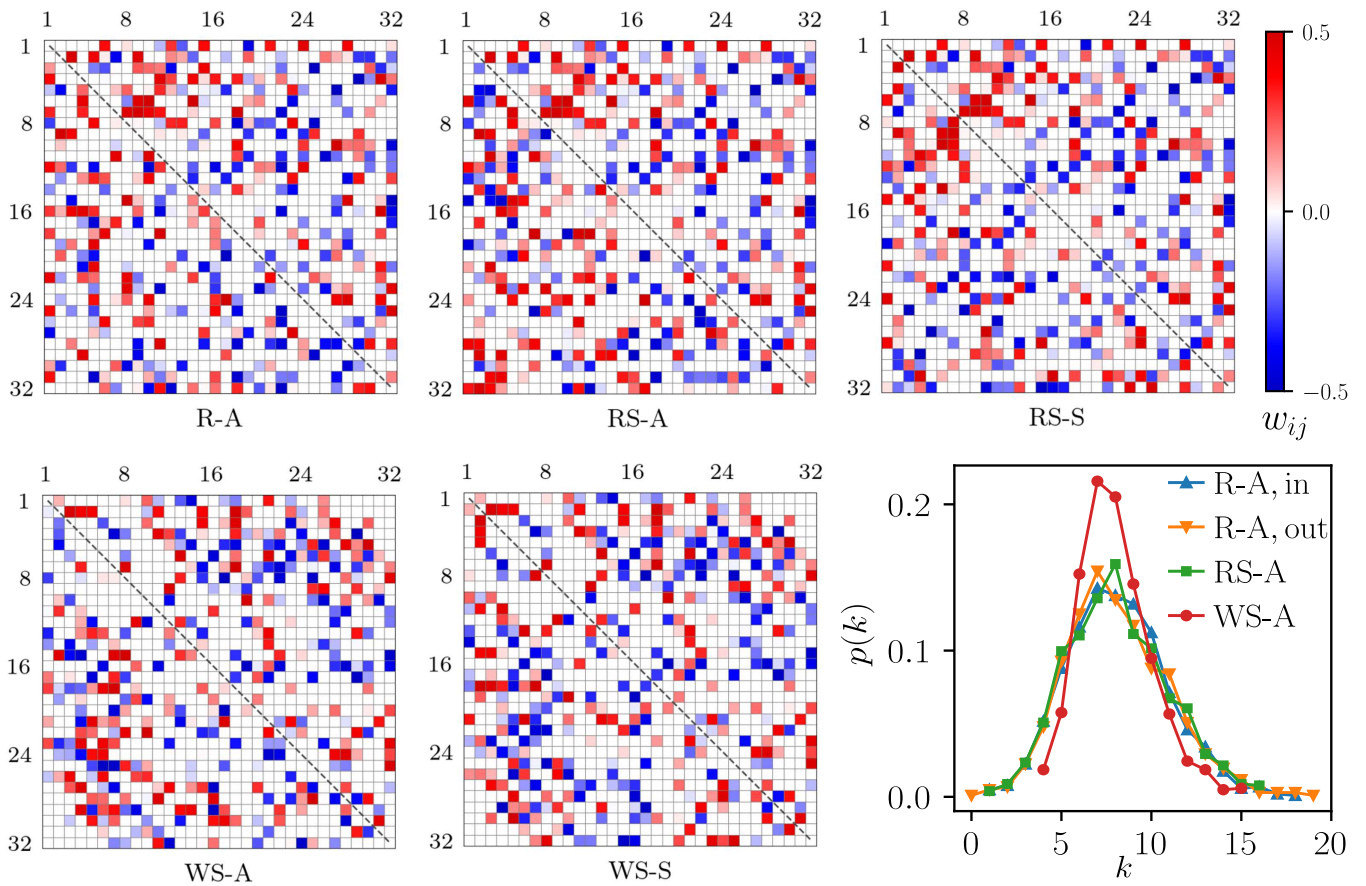


FIG. 1. Reservoir matrices W for five different random network topologies, namely, random-asymmetric (R-A), random symmetrized-asymmetric (RS-A), random symmetrized-symmetric (RS-S), Watts–Strogatz-asymmetric (WS-A), and Watts–Strogatz-symmetric (WS-S). We display examples with $N = 32$ reservoir nodes for a better visibility. In the subsequent analysis of the dynamical systems, $N = 1024$ in most cases. A filled square at position (i, j) stands for an active connection from node i to node j , i.e., $i \rightarrow j$, one at (j, i) for $j \rightarrow i$. The color of each square stands for the weight value, i.e., $w_{ij} \in \mathbb{R}$ for $i \rightarrow j$ and $w_{ji} \in \mathbb{R}$ for $j \rightarrow i$ (see colorbar to the right). The diagonal is indicated in all plots for better visual guidance with respect to symmetry. Bottom-right: node-degree distribution of network topologies R-A, RS-A, and WS-A with 1024 nodes each. The distributions of incoming and outgoing degrees (in, out) are different for asymmetric connection network R-A while they are same for remaining symmetric connection network topologies. The networks RS-S (WS-S) and RS-A (WS-A) are not shown as they have the same node-degree distribution owing to shared connection matrices.

system state information is given to the RC model. In the case of (ii), the RC model requires enough memory for a delay embedding to reconstruct the full state.^{45–48}

We will focus to the latter RC tasks with partial information as input, investigate the performance and its relation with the reservoir topology. The DPTs of multi-dimensional states, namely, Lorenz model (L63), extended Lorenz model (L8) and Galerkin shear flow model (SF), proceed by feeding only 1, 2, and 5 inputs $\mathbf{u}$, respectively, to predict states $\mathbf{y}^p$ of dimensions 3, 8, and 9, respectively, including the ones that are not seen by the RC, see Table I. Finally, we denote a task as *direct prediction* if component u_d predicts component y_d^p and as *cross prediction* if u_d predicts $y_{d'}^p$, i.e., when component indices are different, $d \neq d'$. This implies that open-loop RC tasks with $m < M$ always include cross-prediction tasks.

The five investigated reservoir topologies are summarized in Fig. 1 for a small example network of 32 neurons (to visualize different topologies clearly). They are constructed such that the connections and their weights can be modified independently. This is achieved by taking the quadratic reservoir matrix W as a Hadamard product, which is an element-wise multiplication, of a connection matrix A and a weight matrix W^c , i.e., $W = A \odot W^c$. A similar setup was chosen in Ref. 33. For nodes with indices i and j , the fixed network connections, $i \rightarrow j$ and $j \rightarrow i$, are encoded in the connection matrix A , and their weights, w_{ij} and w_{ji} , in the weight matrix W^c . The weights $w_{ij} \in \mathbb{R}$ are sampled from a uniform distribution, $w_{ij} \sim \mathcal{U}([-0.5, 0.5])$. Thus, $W_{ij} = A_{ij}w_{ij}$. The spectral radius of the reservoir matrix W is then set to a value ρ by normalizing W . The connection matrix A is symmetric if $A = A^T$; otherwise, it is asymmetric. For the weight matrix W^c , symmetry and asymmetry are defined in the same way.

The most general topology is a random-asymmetric network (R-A) with A and W^c both asymmetric, which implies unidirectional connections, $W_{ij} \neq 0$, but $W_{ji} = 0$, along with bidirectional connections with $W_{ij} \neq 0$ and $W_{ji} \neq 0$. For example, nodes 1 and 6 have a unidirectional connection, while nodes 2 and 5 are bidirectionally connected in Fig. 1. The second topology is a symmetrized connection matrix A in combination with an asymmetric weight matrix W^c , i.e., RS-A. The third topology, RS-S, is obtained by taking both A and W^c symmetric, i.e., any pair of nodes is bidirectionally connected. Finally, a (random) Watts–Strogatz (WS) network²³ with rewiring probability $p = 1$ is used, which has by definition a symmetric A , but can obey either an asymmetric or a symmetric weight matrix W^c . This defines WS-A and WS-S in Fig. 1. Here, we do not consider rewiring probabilities $p < 1$, which lead to small-world networks.

The realization of these networks starts with the similar distribution of W_{ij} , called $f(W_{ij})$, in the reservoir matrix W , as shown in the inset of Fig. 2. The distribution $f(W_{ij})$ of the functional reservoir matrix W after setting the spectral radius $\rho = 1$ for all network topologies is shown in the main panel of Fig. 2. The peak in the distribution corresponds to a large number of null entries in W , owing to the small reservoir density of $D_r = 0.008$ which is constant throughout the study. This hyperparameter is chosen very small, which causes sparse reservoir networks. Such a choice of small D_r is supported by studies^{49,50} that demonstrated a high compressibility of the neuron correlations in the human brain.

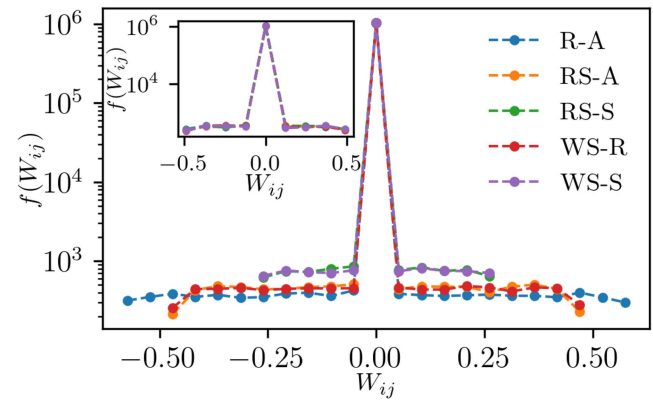


FIG. 2. A representative weight distribution $f(W_{ij})$ of reservoir matrix W for the five network configurations, namely, R-A, RS-A, RS-S, WS-A and WS-S with $N = 1024$ nodes and reservoir density $D_r = 0.008$ after setting the spectral radius (here, $\rho = 1$). Inset: The weight distributions of the same networks before setting the spectral radius. The pronounced peak at $W_{ij} = 0$ is due to small reservoir density $D_r = 0.008$.

III. DYNAMICAL SYSTEMS

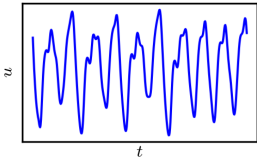
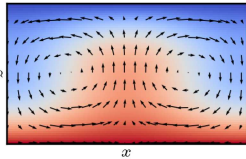
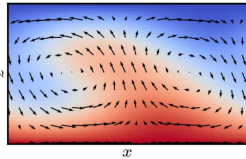
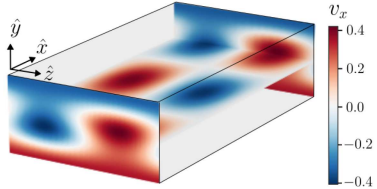
The dynamical systems and their configurations of interest for processing via RC are summarized in Table I. These are the Mackey–Glass (MG) system,⁴³ Lorenz 63 system (L63) with three degrees of freedom,⁴⁰ the extended Lorenz system (L8) with eight degrees of freedom,⁴¹ and a Galerkin model for a three-dimensional plane shear flow (SF) with nine degrees of freedom.⁴² The number of degrees of freedom is given by N_{DoF} in the following. The models L63 and L8 can be derived from a thermal convection flow. The prediction of these systems is performed in open-loop operation of the RC model wherein one-step ahead prediction is realized by feeding the input regularly at each step. Moreover, the fed input is partial with $N_{\text{in}} < N_{\text{out}}$ dynamical variables, where N_{out} defines the number of target states to predict. Here, $N_{\text{out}} = M = N_{\text{DoF}}$, see again Sec. II.

All of these systems show chaotic dynamics, with a positive maximal Lyapunov exponent λ_{max} , in their state space. The N_{DoF} exponents form the Lyapunov spectrum and are ordered, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{N_{\text{DoF}}}$. The Kaplan–Yorke dimension D_{KY} provides an estimate of the dimension of the attractor.⁵¹ It is given by

$$D_{\text{KY}} = s - \sum_{r=1}^s \frac{\lambda_r}{\lambda_{s+1}}, \quad (3)$$

with r th Lyapunov exponent λ_r in the Lyapunov spectrum, such that $\sum_{r=1}^s \lambda_r \geq 0$ and $\sum_{r=1}^{s+1} \lambda_r < 0$. The Lyapunov spectrum of a dynamical system is obtained by $\lambda_r = \frac{1}{K} \sum_{k=1}^K \ln G_r^k$, where G_r^k is the growth rate of the r th unit vector in the orthogonal basis of the state space of the dynamical system at time step k , and the total number of time steps K is large. The growth rates G_r^k for $r = 1, 2, \dots, N_{\text{DoF}}$ are computed using QR-decomposition of the Jacobian matrix J^k , which is obtained by simultaneously evolving the state of the system and the dynamics of the corresponding tangent space from an initial

TABLE I. Summary of the dynamical systems together with their number of degrees of freedom N_{DoF} , the numbers of inputs N_{in} and outputs N_{out} , as well as their maximal Lyapunov exponent λ_{max} and Kaplan–Yorke dimension D_{KY} . We also list the integration time steps width δt and the prediction time step width of the RC, Δt . These are, from left to right, the Mackey–Glass (MG) equation given by Eq. (4), the Lorenz 63 and eight-dimensional Lorenz-type models (L8), which are given by Eq. (B1), and 9-dimensional Galerkin model for a three-dimensional plane shear flow (SF), which is given by Eq. (A1). All prediction tasks for the temporal dynamics of these nonlinear models are performed in the open-loop reservoir computing scenario with the listed input and output degrees of freedom (DoF). In the last row, we display a time series (MG) or a snapshot at a time instant (L63, L8, SF) that shows the flow structures. In L63 and L8, these are the temperature field (colored background contours) together with the velocity vector field. In the case of SF, these are contours in three planes, displaying the three-dimensional spatial structure of the streamwise velocity field component.

Mackey–Glass equation (MG)	Lorenz 63 model (L63)	Lorenz-type model (L8)	Shear flow model (SF)
$N_{\text{DoF}} = \infty$	$N_{\text{DoF}} = 3$	$N_{\text{DoF}} = 8$	$N_{\text{DoF}} = 9$
$N_{\text{in}} = 1$	$N_{\text{in}} = 1$	$N_{\text{in}} = 2$	$N_{\text{in}} = 5$
(u)	(A_1 or B_1)	(A_1, B_1)	(a_1, a_2, a_3, a_4, a_6)
$N_{\text{out}} = 1$	$N_{\text{out}} = 3$	$N_{\text{out}} = 8$	$N_{\text{out}} = 9$
(u)	(A_1, B_1, B_2)	($A_1, \dots, A_4, B_1, \dots, B_4$)	($a_1, \dots, a_9$)
$\lambda_{\text{max}} = 0.006$	$\lambda_{\text{max}} = 0.91$	$\lambda_{\text{max}} = 1.48$	$\lambda_{\text{max}} = 0.02$
$D_{\text{KY}} = 2.10$	$D_{\text{KY}} = 2.06$	$D_{\text{KY}} = 3.44$	$D_{\text{KY}} = 6.25$
$\delta t = 0.017$	$\delta t = 0.002$	$\delta t = 0.002$	$\delta t = 0.002$
$\Delta t = 1$	$\Delta t = 0.05$	$\Delta t = 0.1$	$\Delta t = 1$
			

condition for the k time steps to iterate the computation of f^k at each time step k .⁵²

For SF, D_{KY} is computed within the turbulent regime of the shear flow, well before the dynamics becomes laminar eventually. In the following, we will detail all four models for completeness. We see that D_{KY} increases from L63 to SF, indicating the increasing complexity of the dynamics in the state space.

A. Mackey–Glass equation

The first system we investigate is the nonlinear time-delayed Mackey–Glass (MG) equation⁴³ and is given by

$$\frac{du}{dt} = a \frac{u(t - \tau)}{1 + u(t - \tau)^q} - bu(t). \quad (4)$$

We use standard parameters $a = 0.2$, $b = 0.1$, $q = 10$, and the time delay $\tau = 17$. The dynamics described by time-delayed equation (4) for the parameters, which is chaotic in an attractor with box counting dimension $2 < D < 3$,⁵³ happens in an infinite-dimensional state space. Although the MG equation represents only a scalar function $u(t)$, it is highly nonlinear and exhibits high dynamical variability that depends on the delay parameter τ .

The Mackey–Glass time series⁴³ is obtained by iterating the discrete approximation of the delay differential Eq. (4) following Ref. 54. The initial conditions are sampled from a random uniform distribution, and the first 250 000 iterations are discarded as an initial transient. The time step of the Grassberger–Procaccia iteration is $\delta t = 1.7 \times 10^{-2}$. Further, training and testing data sets are prepared by sampling the time series with a step size of $\Delta t = 1$ and rescaling

them such that $u(t) \in [-1, 1]$. The task is to make a one-step ahead direct prediction of MG time series using RC, see also Table I.

B. Lorenz 63 model and extended eight-dimensional Lorenz model

The standard three-dimensional Lorenz model, which is often referred to as the Lorenz 63 model⁴⁰ as well as its extension to eight dimensions,^{41,55} can be derived from the equations for buoyancy-driven two-dimensional Rayleigh–Bénard convection flow between two impermeable parallel plates at distance H .⁵⁶ The configuration together with the corresponding boundary conditions is shown in Fig. 3. A fluid flow layer between two impermeable plates is heated uniformly from below and cooled from above. The top plate is kept at a lower temperature T_H than the bottom plate, which is at $T = T_0$. Thus, $\Delta T = T_0 - T_H > 0$. At the top and bottom, free-slip boundary conditions (see the gray box to the right in the figure) are set for the two-dimensional incompressible velocity vector field $\mathbf{u} = [u_x(x, z, t), u_z(x, z, t)]^T$. In the x -direction, periodic boundary conditions are applied. The equilibrium state of the convection layer is given by $\mathbf{u} = 0$ and $T_{\text{eq}}(z) = T_0 - z\Delta T/H$. Once the temperature difference ΔT exceeds a critical threshold value, fluid motion in the form of the sketched circulation rolls starts by an onset of a linear flow instability. The (turbulent) fluid motion is driven by buoyancy forces.

All Lorenz-type models are obtained by spatial mode expansions of the scalar velocity stream function $\zeta(x, z, t)$, which substitutes the incompressible two-dimensional velocity vector field, and the temperature fluctuation field $\theta(x, z, t) = T(x, z, t) - T_{\text{eq}}(z)$.

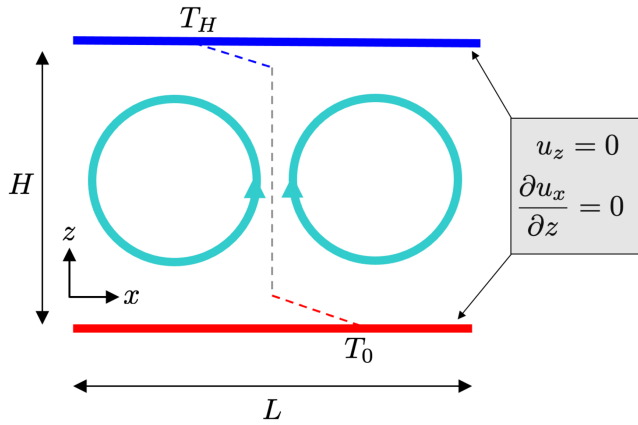


FIG. 3. Sketch of the two-dimensional Rayleigh–Bénard convection configuration for both Lorenz models. We indicate the boundary conditions and the counter-rotating flow circulations in the form of convection rolls in the middle of the layer.

These expansions satisfy the boundary conditions, which we outlined in the caption of Fig. 3. The series expansions are given by

$$\zeta(x, z, t) = \sum_{i,j=1}^{\infty} c_{\zeta} A_{ij}(t) \Phi(\alpha_i x) \sin(\beta_j z), \quad (5)$$

$$\theta(x, z, t) = \sum_{k,l=1}^{\infty} c_{\theta} B_{kl}(t) \Phi(\alpha_k x) \sin(\beta_l z), \quad (6)$$

with normalization prefactors c_{ψ}, c_{θ} , the real amplitudes $\{A_{ij}(t), B_{kl}(t)\}$, and the wavenumbers $\alpha_k = k\pi/L$ and $\beta_k = k\pi/H$. In the extended Lorenz model, truncation is done after the fourth term in both expansions, resulting in eight modes. We take the critical wavelength of the primary linear convective instability as the side length L of the domain. The aspect ratio is $\Gamma = L/H = 2\sqrt{2}$. This leads to a geometry factor $b = 8/3$ in the Lorenz models.

The four selected velocity modes are $A_1 = A_{11}, A_2 = A_{01}, A_3 = A_{12}$, and $A_4 = A_{03}$; the four selected temperature modes are $B_1 = B_{11}, B_2 = B_{02}, B_3 = B_{12}$, and $B_4 = B_{04}$ in accordance with the notation in Eqs. (5) and (6). The set of coupled nonlinear ordinary differential equations of our eight-dimensional Lorenz system (L8) is formally given by

$$\frac{dA_i}{dt} = F_i^{(A)}(A_j, B_k; \sigma, b, r), \quad (7)$$

$$\frac{dB_i}{dt} = F_i^{(B)}(A_j, B_k; \sigma, b, r), \quad (8)$$

with $i, j, k = 1, \dots, 4$ and system parameters $\sigma = 10, b = 8/3$, and $r = 28$. The detailed equations are listed in Appendix B 1. This model is integrated numerically with a fourth-order Runge–Kutta scheme and the obtained time series of the expansion coefficients are used as input as well as ground truth in the reservoir computing

experiments. The integration time step is $\delta t = 2 \times 10^{-3}$. The initial conditions are randomly chosen, and a long enough transient at the beginning is discarded. The additional higher-order modes with their smaller amplitudes lead to a conservation of total energy E and vorticity Ω . Both invariants are given by

$$E(t) = \frac{1}{2A} \int_A ((\nabla \zeta)^2 - z\theta) dA \quad \text{and} \quad \Omega(t) = \frac{1}{A} \int_A \omega dA, \quad (9)$$

with the vorticity $\omega = -\nabla^2 \zeta$ and the convection domain size $A = LH$.

The well-known Lorenz 63 model includes three of the eight coefficients only, A_1, B_1 , and B_2 .⁴⁰ The data for the L63 model are also generated similar to the L8 model with the same integration time step of $\delta t = 2 \times 10^{-3}$. The flow constructed from the modes of the extended Lorenz model describes the shear motions, as visible by the tilt in velocity and temperature fields. A snapshot of the convection flow dynamics for L63 and L8 is shown in Table I.

C. Galerkin model for plane shear flow

The third dynamical system, which we use in our investigation, is a nine-mode Galerkin model^{42,57} of a simple shear flow, sometimes also known as the minimal flow unit.⁵⁸ This three-dimensional nonlinear model describes the elementary dynamical cycle of near-wall coherent flow structures present in every wall-bounded turbulent flow. The present model is for a shear flow between parallel walls at distance d_0 with free-slip boundary conditions for the velocity field subject to a sinusoidal body force which is added to the Navier–Stokes equations to sustain the sinusoidal shear flow⁴² in the domain of $0 \leq x \leq L_x, -1 \leq y \leq 1$, and $0 \leq z \leq L_z$. The starting point is the expansion of velocity field $\mathbf{v}(x, y, z, t) = (v_x, v_y, v_z)$ as

$$\mathbf{v}(x, y, z, t) = \sum_{i=1}^9 a_i(t) \mathbf{v}_i(x, y, z), \quad (10)$$

with prescribed spatial modes $\mathbf{v}_i$, which all satisfy the free-slip boundary conditions with respect to the y or wall-normal direction and periodicity in the other two space directions. The detailed spatial structure of the modes is found in Ref. 42. The modes considered are the basic sinusoidal shear flow profile $\mathbf{v}_1(y) = \sqrt{2} \sin(\pi y/2) \mathbf{e}_x$, the streamwise streak $\mathbf{v}_2$, the downstream vortex $\mathbf{v}_3$, the spanwise flow modes $\mathbf{v}_4$ and $\mathbf{v}_5$, the normal vortex modes $\mathbf{v}_6, \mathbf{v}_7$, and finally the modification of the basic velocity profile due to turbulence $\mathbf{v}_9$. We take $L_x = 4\pi$ and $L_z = 2\pi$.

The dynamics of these modes is obtained by a projection of the three-dimensional Navier–Stokes dynamics on each of the nine modes. This results in a coupled and closed set of nonlinear ordinary differential equations for the expansion coefficients $a_i(t)$, see Eq. (10), when higher-order nonlinearities are truncated. The time evolution of the coefficients $a_i(t)$ with $i = 1, \dots, 9$ in our dynamical system SF follows to

$$\frac{da_i}{dt} = F_i(a_j; \alpha, \beta, \gamma, \text{Re}) \quad \text{with} \quad i, j = 1, \dots, 9, \quad (11)$$

and system parameters $\alpha = 2\pi/L_x, \beta = \pi/2$, and $\gamma = 2\pi/L_z$. The detailed system is given in Appendix A 1. The central parameter in the dynamical model is the Reynolds number $\text{Re} = U_0 d_0 / (2\nu)$.

Here, U_0 is the characteristic amplitude of the basic flow and ν is the kinematic viscosity of working fluid. Table I shows a snapshot of the spatial structure of the streamwise velocity component v_x reconstructed from ansatz (10) displaying the typical streamwise streaky structures.

The data are generated by numerical integration of the SF model (11) for $Re = 500$ using a Runge–Kutta–Fehlberg (RK45) scheme with initial conditions selected randomly on an energy shell $\sum_{i=1}^9 a_i^2 = 0.1$. The integration time step is 2×10^{-3} . This completes the description of the four dynamical systems which we study in the following.

IV. RESULTS

In the following, we compare the performance of the five different reservoir computing models, each applied to the prediction tasks in four different nonlinear dynamical systems, MG, L63, L8, and SF, which we described in detail in Sec. III. The input data for all the systems are rescaled to an interval of $[-1, 1]$. The five RC models differ in the topology of their random reservoirs, which we have detailed in Sec. II, namely, the reservoir topologies R-A, RS-A, RS-S, WS-A, and WS-S. The statistical analysis is performed as follows:

- (1) In total, we have five hyperparameters that can be varied in the present RC study. These are the number of neurons N , the reservoir density D_r , the spectral radius $\rho(W)$, the leaking rate ε , and the Tikhonov regularization parameter γ . We keep N and D_r fixed; the grid search is thus done with respect to $(\rho, \varepsilon, \gamma)$. The neuron number is varied only for the reservoir size dependence study. At the end of Sec. II, we already motivated the low value of D_r . Note that for each specific task in the following, this hyperparameter optimization has to be done individually.
- (2) The workflow is as follows. For each individual open-loop learning task with a selected dynamical system and reservoir network topology, we perform a hyperparameter tuning (as just described) for each chosen prediction time step Δt to obtain the optimal one. Recall that W^{in} is fixed initially and left unchanged. After this training, we analyze the performance in the testing phase by the mean squared error (MSE). The statistics of each specific learning task implies that:
 - Each hyperparameter grid search point at a fixed Δt is conducted for 100 individual random realizations of the reservoir matrix W . This training phase includes 2000 training steps. The finally chosen hyperparameters correspond to the median of the minimum MSEs from these 100 realizations of W . The MSE is computed over 2000 prediction steps. The grid-search is done for $\rho \in [0.0, 1.5]$, $\varepsilon \in [0.1, 1.0]$ and $\gamma \in [10^{-13}, 10^0]$ (logarithmically in steps of 10). The hyperparameter search is thus conducted on a uniform grid of size $16 \times 10 \times 14$ points.
 - The subsequent computation of the MSE at this fixed Δt is done for 50 individual random realizations of the reservoir matrix W . This testing phase includes 2000 test steps.

- The computation of the MSE for optimal hyperparameters at optimal prediction time step Δt includes 1000 individual random realizations of W . These testing phases comprise again 2000 steps.

We reiterate that this procedure is carried out for each learning task, including the tests presented in Appendices C and D.

A. Optimal reservoir computing time step

The reservoir update proceeds in correspondence with Eq. (1) with a time step Δt to advance from time instant k to the next, $k + 1$ in this notation. This time step Δt defines the sampling time step and is a multiple of the actual numerical integration time step of the dynamical system, the latter of which we denoted by δt , see again Table I. RC models respond differently to the sampling of input data as shown in Ref. 47. Thus, each RC model in combination with the underlying task has a characteristic time step to perform the computation optimally. To this end, the reservoir computing time step, i.e., the sampling interval of input data Δt , is determined first, by optimizing the median MSE, in Eq. (12), of each DPT over an ensemble of 50 randomly initialized reservoirs for each topology. The number of neurons has been fixed to $N = 1024$ for this pre-analysis. The DPTs correspond to those, which are indicated in Table I.

Figure 4 summarizes the results on the reservoir computing time step for open-loop scenario with $N_{in} < N_{out}$ (see again Table I). The integration time step for DPTs, namely, L63, L8, and SF, is $\delta t = 0.002$. The data is sampled at an interval of $\Delta t = 25\delta t = 0.05$ for L63, $\Delta t = 50\delta t = 0.1$ for L8, and $\Delta t = 2500\delta t = 1$ for SF. It is observed that there is practically no influence of reservoir topology on the MSE in the cases of L63 and SF, see Figs. 4(b) and 4(d). For the eight-dimensional extended Lorenz model, a dependence of the MSE on the topology is observed in Fig. 4(c). The optimal reservoir computing time step remains practically the same for all reservoir topologies. The manifestation of the optimal sampling time step is illustrated in Fig. 5, which shows the attractor of L63 with modes A_1 and B_1 reconstructed by RC with input mode B_1 for $\Delta t = 0.01$ and the optimal $\Delta t = 0.05$. The reconstruction of the attractor with input data B_1 sampled at $\Delta t = 0.05$ is visually better than that at $\Delta t = 0.01$, although the former is coarser and no longer smoother compared to the latter. This indicates that RC learns better with coarser input data and enables longer prediction.

Finally, in the case of the MG system, the smaller the reservoir computing time step Δt , the better the prediction as shown in Fig. 4(a). This is because the task becomes more linear as the prediction horizon is decreased (which is not the case for the tasks involving cross-prediction). We work for the following analysis with $\Delta t = 1$. After preparing the input data, we can now proceed to a detailed performance analysis in the next Subsection.

B. Performance for dynamics prediction task

In the following, we present the performance of all reservoir network topologies for their optimal hyperparameter sets. All DPTs, which are carried out using reservoirs of $N = 1024$ nodes with a reservoir density of $D_r = 0.008$, are thoroughly explored and discussed in detail. In addition, we consider smaller reservoir sizes with $N = 256$ and 512 nodes for comparison. The MSE between

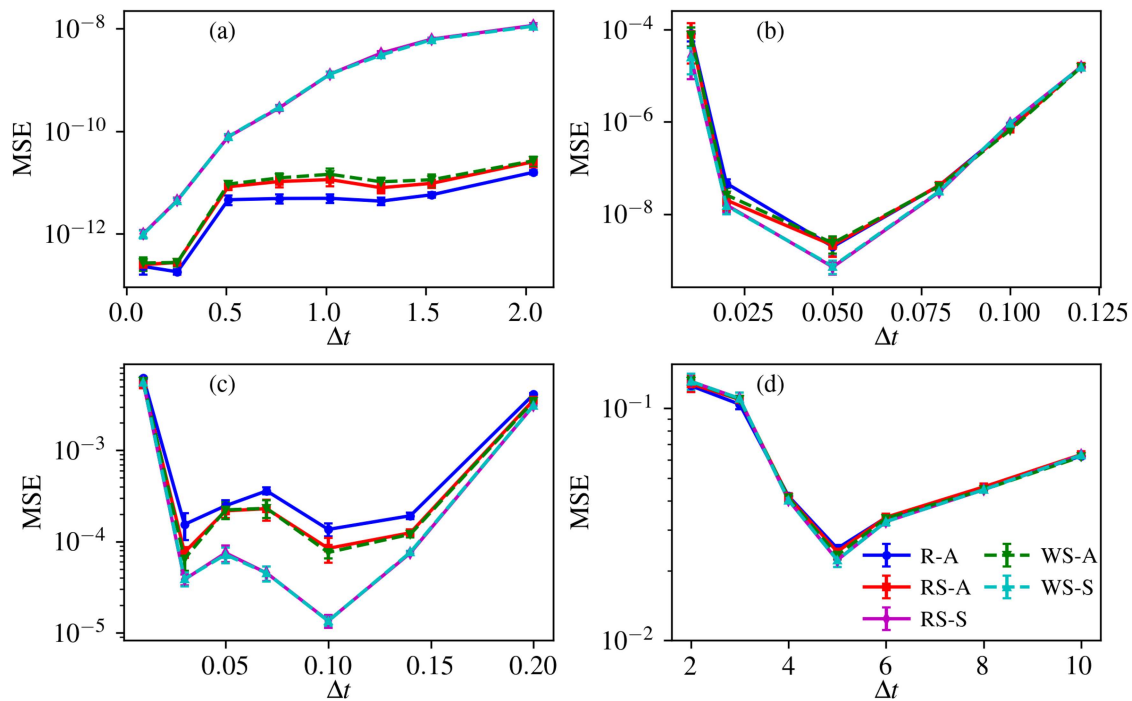


FIG. 4. Optimal time step size Δt for sampling the input data of the (a) Mackey–Glass equation (MG), (b) Lorenz 63 (L63) model, (c) eight-dimensional Lorenz-type (L8) model, and (d) Galerkin model of the three-dimensional plane shear flow (SF) for reservoir networks with node numbers $N = 1024$. Each case for each application, and each topology is obtained as an ensemble median of 50 different random reservoir network initializations. The legend in (d) applies to all panels.

the ground truth vector $\mathbf{y}(k) = [y_1, y_2, \dots, y_{N_{\text{out}}}]^T$ and the predicted vector $\mathbf{y}^p(k) = [y_1^p, y_2^p, \dots, y_{N_{\text{out}}}^p]^T$ is obtained by

$$\text{MSE} = \frac{1}{k_p - k_t} \sum_{k=k_t+1}^{k_p} \|\mathbf{y}(k) - \mathbf{y}^p(k)\|^2, \quad (12)$$

where $\|\cdot\|$ denotes L^2 norm. This can be written as the sum of the MSEs of all N_{out} tasks of a DPT, i.e.,

$$\text{MSE} = \sum_{d=1}^{N_{\text{out}}} \text{MSE}_d, \quad (13)$$

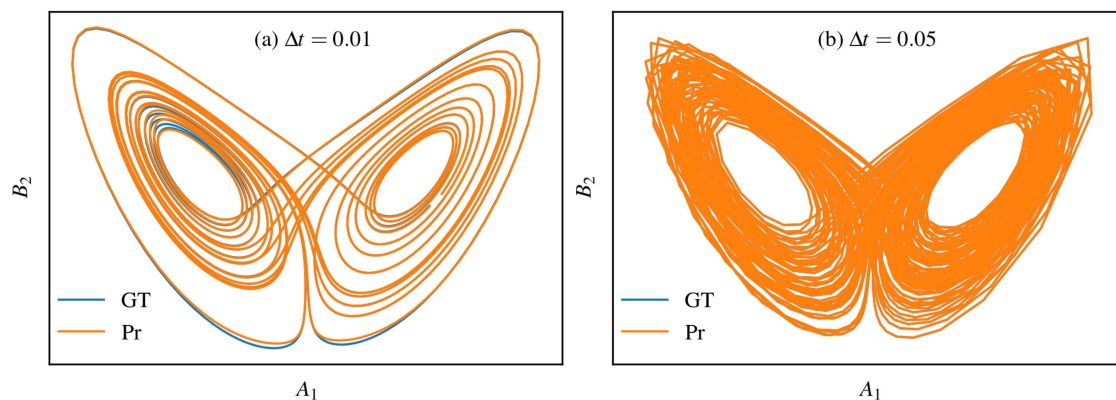


FIG. 5. Reconstruction of the attractor of the Lorenz 63 system with respect to output modes A_1 and B_2 for reservoir computing time step (a) $\Delta t = 0.01$ and (b) $\Delta t = 0.05$. The input mode is B_1 . Solid blue line shows ground truth (GT) of A_1 and B_1 , and solid orange line represents their prediction (Pr).

TABLE II. Comparison of the optimal reservoir computing model hyperparameters spectral radius ρ and Tikhonov regularization parameter γ with leaking rate $\varepsilon = 0.7$ as obtained from a grid search in an open-loop prediction scenario. Four learning tasks, Mackey–Glass equation (MG), three- and eight-dimensional Lorenz model (L63, L8), and nine-dimensional Galerkin model for a three-dimensional shear flow (SF) are analyzed. Five different random network topologies of the reservoir are used for each case. The reservoir matrix W has $N = 1024$ nodes and $D_r = 0.008$.

	MG (ρ, γ)	L63 (ρ, γ)	L8 (ρ, γ)	SF (ρ, γ)
R-A	(1.2, 10^{-11})	(0.7, 10^{-11})	(0.5, 10^{-11})	(0.9, 10^{-1})
RS-A	(1.3, 10^{-11})	(0.8, 10^{-11})	(0.9, 10^{-10})	(1.1, 10^{-2})
RS-S	(1.2, 10^{-11})	(1.0, 10^{-11})	(1.0, 10^{-11})	(1.0, 10^{-2})
WS-A	(1.3, 10^{-11})	(0.8, 10^{-11})	(0.9, 10^{-10})	(1.1, 10^{-2})
WS-S	(1.2, 10^{-11})	(1.0, 10^{-11})	(1.0, 10^{-11})	(1.0, 10^{-2})

where MSE_d is the prediction error of d th task (or component), which is defined as

$$\text{MSE}_d = \frac{1}{k_p - k_t} \sum_{k=k_t+1}^{k_p} (y_d(k) - \hat{y}_d(k))^2. \quad (14)$$

The analysis of each DPT is performed over an ensemble of 10^3 different realizations of the reservoir coupling matrix with fixed input weights for all simulations with a given number of nodes N .⁵⁹ We also analyze the relative improvement of the performance of the MSE for a specific DPT with respect to the network topology that performs the *worst* with mean squared error MSE^w for the same DPT. This measure is given by

$$I_{\text{err}} = \frac{|\text{MSE}^w - \text{MSE}|}{\text{MSE}^w}. \quad (15)$$

For the three-dimensional shear flow (SF), we list in addition the normalized relative error (NARE) (in %), which is defined in Appendix A in Eq. (A2) and listed in Table IV. This measure is useful to test the success of the RC model for predicting statistical correlations of the velocity field of the shear flow. Since it is of primarily fluid mechanical interest, it has been moved into the Appendix.

The grid-search is performed in hyperparameter space defined by varying the spectral radius, the leaking rate and the Tikhonov regularization parameter as listed at the beginning of Sec. IV. Note that the optimal leaking rate in the grid search is $\varepsilon = 0.7$ for all tasks. Consequently, this hyperparameter will be practically also constant

TABLE III. Performance in an open-loop scenario for the Mackey–Glass (MG) equation, three- and eight-dimensional Lorenz-type model (L63, L8), and nine-dimensional Galerkin model of a three-dimensional shear flow (SF). For all cases, we provide the MSE with error bars. The performance improvement I_{err} as defined in Eq. (15) is specified in parentheses [·], given in per cent.

	MSE-MG [I_{err} %]	MSE-L63 [I_{err} %]	MSE-L8 [I_{err} %]	MSE-SF [I_{err} %]
R-A	(5 ± 1) × 10⁻¹² [99.6]	(3 ± 1) × 10 ⁻⁹ [25.0]	(1.5 ± 0.2) × 10 ⁻⁴ [0.0]	(2.5 ± 0.1) × 10 ⁻² [3.8]
RS-A	(1.6 ± 0.5) × 10 ⁻¹¹ [98.8]	(4 ± 2) × 10 ⁻⁹ [0.0]	(1.2 ± 0.4) × 10 ⁻⁴ [20.0]	(2.6 ± 0.2) × 10 ⁻² [0.0]
RS-S	(1.3 ± 0.2) × 10 ⁻⁹ [0.0]	(8 ± 3) × 10⁻¹⁰ [80.0]	(1.3 ± 0.2) × 10⁻⁵ [91.3]	(2.4 ± 0.1) × 10⁻² [7.7]
WS-A	(1.6 ± 0.5) × 10 ⁻¹¹ [98.8]	(3 ± 2) × 10 ⁻⁹ [25.0]	(1.2 ± 0.4) × 10 ⁻⁴ [20.0]	(2.6 ± 0.2) × 10 ⁻² [0.0]
WS-S	(1.3 ± 0.2) × 10 ⁻⁹ [0.0]	(8 ± 3) × 10⁻¹⁰ [80.0]	(1.3 ± 0.2) × 10⁻⁵ [91.3]	(2.4 ± 0.1) × 10⁻² [7.7]

throughout the study. The optimal hyperparameters for $N = 1024$ are listed in Table II for each DPT.

The MSE results of the DPTs are summarized in Table III. The smallest MSE values are always indicated in bold. For the MG case with $N_{\text{out}} = 1$, the random network with R-A topology predicts the best among all the networks studied, which confirms the results of Ref. 33. The prediction from the networks RS-A and WS-A, which are similar in connection and weight distribution, are comparable. Similar results follow for networks RS-S and WS-S. For the cases of L63 with $N_{\text{out}} = 3$ and L8 with $N_{\text{out}} = 8$, it is the symmetric reservoir topology for, both connection and weights (namely, RS-S and WS-S), that performs best among the five reservoir topologies. For the SF case with $N_{\text{out}} = 9$, the performance difference between RS-A (WS-A) and RS-S (WS-S) is significantly reduced. Thus, we observe a trend that symmetric networks give a better prediction of the dynamics with respect to other network topologies when $N_{\text{in}} < N_{\text{out}}$ for the tasks and networks studied here. The differences seem, however, to decrease with growing dimensionality.

The robustness of this result is further investigated for smaller reservoir sizes of node numbers $N = 256$ in panel (a) and $N = 512$ in panel (b) of Fig. 6. It is seen that the MSE decreases with growing reservoir size for all tasks, even though this trend remains small for SF. The trend is also visible when showing the MSE normalized by maximum MSE, MSE_{max} , for each task. This is displayed in Figs. 6(d)–6(f). In the following, we further investigate the individual tasks and their cumulative effect on the corresponding DPT.

C. Performance for cross-prediction tasks

In the following, we refine the analysis to component-wise cross predictions to explore the dependence of the performance on symmetry or asymmetry. In a cross-prediction task, the component y_d is predicted by training the RC model with inputs $u_{d'}$ and $d \neq d'$. In the case of L63, for example, A_1 is fed in and B_1 is predicted. To compare the performance of d th task with other tasks, the normalized root mean squared error (NRMSE_{*d*}) in Eq. (16) is used, which is given by

$$\text{NRMSE}_d = \sqrt{\frac{\text{MSE}_d}{\text{Var}_d}}, \quad (16)$$

where

$$\text{Var}_d = \frac{1}{k_p - k_t} \sum_{k=k_t+1}^{k_p} (y_d(k) - \langle y_d \rangle)^2 \quad (17)$$

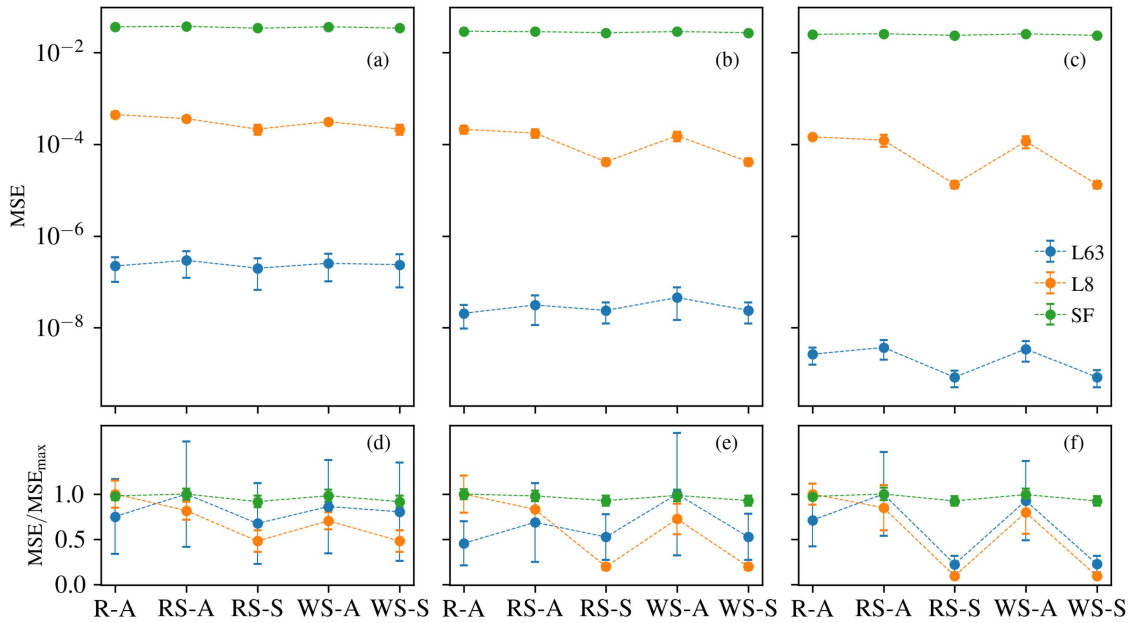


FIG. 6. Comparison of the performance of the five different reservoir computing models from prediction tasks of the Lorenz 63 (L63), the eight-dimensional Lorenz-type (L8), and the Galerkin model of the three-dimensional plane shear flow (SF) for reservoir networks with node numbers (a) 256, (b) 512, (c) 1024. See also Table I for details on the number of inputs and outputs. In panels (a)–(c), we display the mean squared error (MSE). Panels (d)–(f) display the MSE normalized by maximum MSE of each task. Each case for each application, each topology, and each number of reservoirs is obtained as an ensemble median of 10^3 different random reservoir network initializations. The legend in panel (c) applies to all six panels.

is the variance in the ground truth $y_d(k)$ of the task with respect to the temporal mean $\langle y_d \rangle$ during the test phase $k_t < k < k_p$.

The DPT of L63 comprises of three tasks corresponding to the modes A_1, B_1, B_2 (refer to Subsection III B), and the performance of the reservoir topologies for each task is shown in Fig. 7(a). The reservoir state is driven by the input B_1 and output matrix is trained to predict either A_1, B_1 , or B_2 . This is a one-step direct prediction of B_1 and a one-step cross prediction of A_1 or B_2 . We see from the figure that the direct prediction of B_1 performs better

with asymmetric networks. The cross-prediction performance by asymmetric reservoirs is also better for A_1 . However, the symmetric reservoir topology gives better results for the cross prediction of B_2 , and the latter is exactly the degree of freedom, that dominates the MSE of the DPT. Thus, we conclude that symmetric reservoirs display a better cross-prediction performance for L63.

The DPT of L8 consists of eight individual tasks of predicting the modes $A_1, B_1, B_2, A_2, A_3, B_3, A_4, B_4$ in Eq. (8), out of which modes A_1 and B_1 are fed as input to the RC model. Figure 7(b) shows

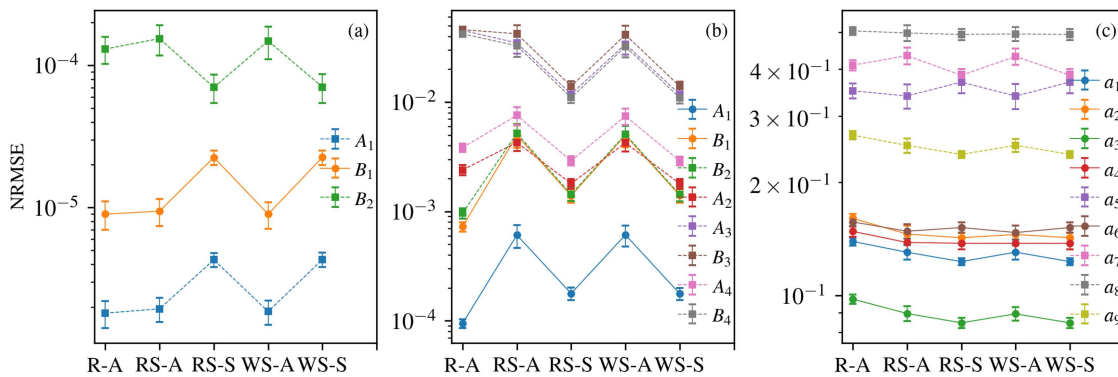


FIG. 7. Task-wise comparison of normalized root mean squared error (NRMSE), see Eq. (16), for the network performance of the DPT of (a) L63, (b) L8, and (c) SF for $N = 1024$ nodes. Solid lines with filled circle marker represent direct-prediction tasks; dashed lines with filled square marker show cross-prediction tasks. See Subsection III B and III C for the legends of the three cases. In L63, the mode B_1 is fed, in L8 the modes A_1 and B_1 , and in SF the five modes given in Table I.

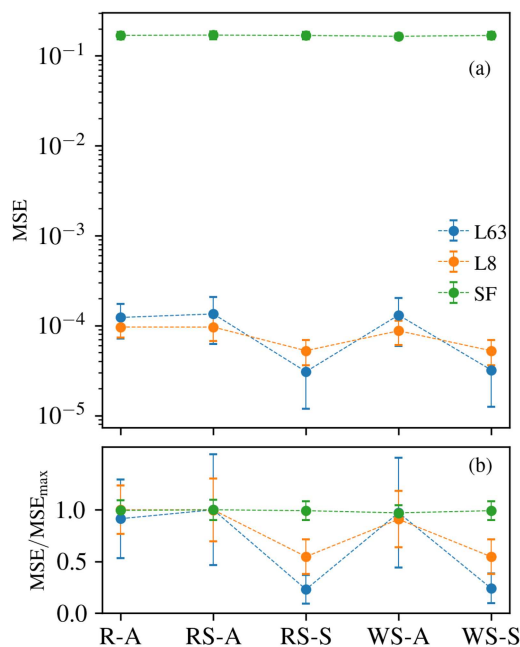


FIG. 8. (a) Comparison of MSE for the network performance of the DPT with sub-optimal time steps $\Delta t = 0.01$ for L63, with $\Delta t = 0.02$ for L8, and $\Delta t = 2.0$ for SF. All reservoirs have $N = 1024$ nodes. (b) MSE normalized by maximum MSE of each task. Each case for each application, each topology, and each number of reservoirs is obtained as an ensemble median of 1000 different random reservoir network initializations. The legend in (a) applies to both panels.

that the direct prediction of A_1 and B_1 performs better with the asymmetric (R-A) reservoir topology, and the cross prediction of A_3, B_3, A_4 , and B_4 gets better by the symmetric reservoir topology. The overall performance of L8, as shown in Figs. 6(c) and 6(f), is thus determined by the magnitude of the NRMSE coming from exactly these cross-prediction tasks. The cross prediction of the remaining

modes A_2 and B_2 performs better by symmetric (RS-S, WS-S) and asymmetric (R-A) reservoir topologies, respectively, for L8.

The differences in performance for the five reservoir topologies are less pronounced for the DPT of SF, which comprises of nine individual tasks, see Fig. 7(c), with respect to both, direct- and cross-prediction tasks. This holds, in particular, for the largest MSE amplitudes. We observe that the dominant cross prediction of a_8 in this DPT is practically insensitive to the reservoir topology. The DPT of the relatively complex SF system with a larger Kaplan–Yorke dimension D_{KY} shows a slightly better performance in the case of a symmetric reservoir topology, see again Fig. 6(f), which we interpret as the cumulative result of the individual prediction performances of modes in Fig. 7(c).

Further, the performance of the networks is investigated for the DPTs with a suboptimal set of reservoir computing time steps $\Delta t = 0.01, 0.02, 2.0$ for dynamical systems L63, L8, and SF, respectively. This analysis underlines the importance of the choice of Δt for the DPT in an open-loop scenario. The results are shown in Fig. 8. The symmetry of the reservoir network topology becomes statistically less significant for suboptimal reservoir computing time steps, cf. Fig. 6(f). The trends in individual tasks of the DPTs, however, remain similar to those with optimal reservoir computing time step. The direct predictions perform, in general, better with an asymmetric network topology—either R-A or RS-A and WS-A—for all the three systems, as shown in Fig. 9. The cross predictions, which dominate the MSEs in Fig. 8, tend to be better processed by RC models with symmetric network topology.

Finally, we examine the performance and topology relation when full state information is provided as the input to the RC model, i.e., in the absence of cross prediction. We consider the dynamics prediction task of the L63 system only. The RC model receives all three time series $[A_1, B_1, B_2]$ as input and the optimum reservoir computing time step changes now to $\Delta t = 0.03$. Asymmetric random networks learn the dynamics better than their symmetric counterparts, irrespective of reservoir computing time step. This is shown in Fig. 10. It was also observed in the prediction of the dynamics described by the Mackey–Glass equation.³³

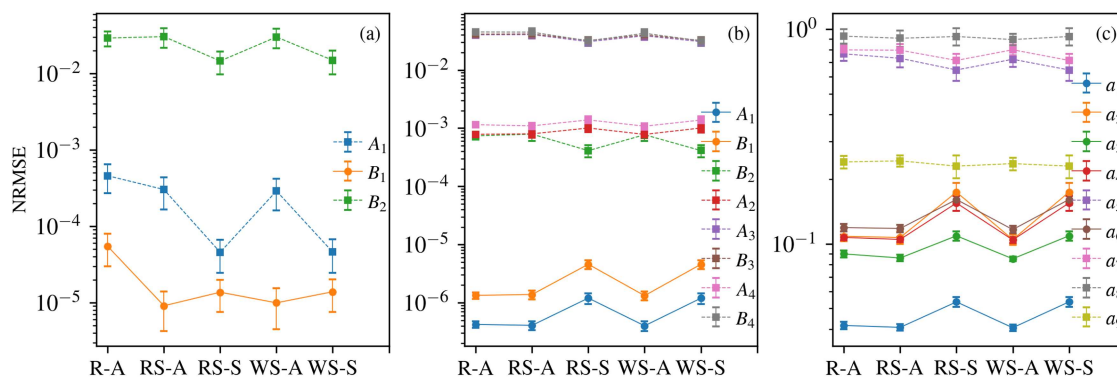


FIG. 9. Task-wise comparison of NRMSE for the network performance of the DPT of (a) L63 with $\Delta t = 0.01$, (b) L8 with $\Delta t = 0.02$, and (c) SF with $\Delta t = 2.0$ for $N = 1024$ nodes. Solid lines with filled circle marker represent direct prediction and dashed lines with filled square marker show cross prediction.

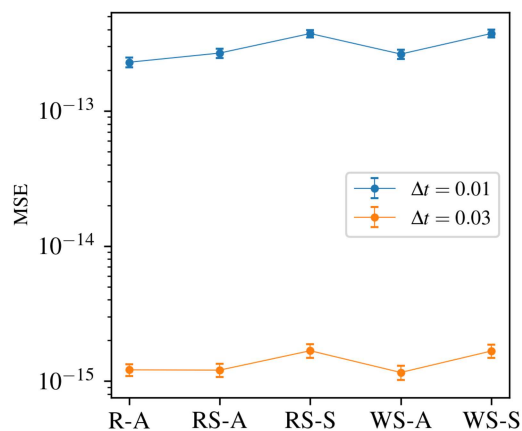


FIG. 10. Comparison of MSE for the network performance of the DPT of L63 with $\Delta t = 0.01$, and $\Delta t = 0.03$ for $N = 1024$ nodes. Here, we apply a direct prediction of all three degrees of freedom.

To summarize our study, the RC model with the optimal sets of hyperparameters in the open-loop scenario perform differently well for the five network topologies and the four selected prediction tasks of increasing dynamical complexity. In Fig. 6, the R-A topology, which we identified in Ref. 33 as the best performing network, is typically not the best reservoir topology for tasks with more degrees of freedom, which involve cross-prediction tasks. The statement, that the asymmetric reservoir learns best, is thus not generalizable and depends on the specifics of the dynamical prediction task. For the tasks with $N_{in} < N_{out}$, due to cross prediction, we find that symmetric networks perform generally better. This can be concluded from the three additional learning tasks next to MG, which are characterized by a quadratic nonlinearity, as present in fluid flow systems.

V. CONCLUSIONS AND OUTLOOK

The first main goal of the present study was to shed new light on the connection between reservoir computing model performance and topology of the central building block of this class of recurrent machine learning algorithms, the sparse random reservoir network. To this end, we set up a systematic construction scheme that can separately prescribe, both the symmetry of network connectivity and weights along the active node connections.

Second, we wanted to carry this study to applications beyond the almost exclusively used academic benchmarks of Mackey–Glass time-delayed and Lorenz 63 model dynamics. Our study thus included an extended Lorenz model, which displays more realistic fluid flow and temperature patterns of two-dimensional thermal convection at higher Rayleigh numbers and a model for the complex near-wall dynamics in a three-dimensional shear flow that is about to transition to turbulence. These models are representative dynamical systems for complex fluid flows, characterized by a quadratic nonlinearity, which arises from the advection term in the

Navier–Stokes equations. A motivation for this analysis is to explore the applicability of RC for fluid flows, either as a surrogate model for the large-scale dynamics in fluid flows or as a subgrid-scale parametrization for unresolvable turbulent processes in coarse-grid simulation models.⁶⁰

It actually turned out, that the two application cases with more degrees of freedom, an extended Lorenz-type model and a three-dimensional shear flow, showed a decreasing susceptibility of the specific reservoir network topology to the learning performance. We suspect, but do not show it, that this trend will continue to other systems, such as the Lorenz 96 model for global circulation processes since the polynomial degree of the nonlinearity remains the same.⁶¹

More detailed, three topologies of networks in RC, that are conceived on the basis of the node connections and their weights, were investigated. These topologies are R-A, RS-A, and RS-S, which are constructed from randomly initialized connection and weight matrices. These topologies, along with two more random network types constructed from the Watts–Strogatz framework with a rewiring probability of $p = 1$, were taken. They were termed WS-A and WS-S. The latter two topologies are similar to RS-A and RS-S, except for a difference in the width of the node-degree distribution.³³ These topologies are used to predict the dynamics of three fluid flow systems, all of which can be described by a Galerkin-type nonlinear dynamical systems model. These reduced-order models prescribe the spatial modes in correspondence with the boundary conditions of the fluid flow problem and describe the dynamics of the expansion coefficients, which get coupled in a system of nonlinear ordinary differential equations. The tasks differ by increasing order of dimensionality (or number of degrees of freedom) and complexity, the latter of which is quantified by the Kaplan–Yorke dimension. These systems are L63, L8, and SF with dimensions 3, 8, and 9, respectively. A one-dimensional time series representing the dynamics from a Mackey–Glass equation at a fixed delay time τ is additionally considered for comparison with the aforementioned systems.

Our open-loop one-step prediction scenario is as follows: the RC predicts the dynamics, which comprises all degrees of freedom of the target nonlinear dynamical system, with partial information at the input of the RC model. We find a generic trend, namely, that the dynamics prediction is performed better if symmetric random network topology (RS-S or WS-S) is used. This holds for the three models that describe two-dimensional Rayleigh–Bénard convection (L63, L8) and three-dimensional shear flow (SF). A disentanglement of the prediction into individual direct- and cross-prediction tasks revealed the following results: (i) the contribution to the total MSE coming from the direct-prediction pipelines is negligible in comparison to those coming from cross predictions; (ii) cross-prediction subtasks thus determine the overall performance. They are the ones, that require a delay-embedding, i.e., a looping of the information inside the network which is a short-term memory. This finding is complemented by several previous works, which have shown that performance can be improved using pre- or post-processing which introduces explicit delay.^{45–48} Here, we observed that exactly these cross-prediction subtasks are better carried out by symmetric networks, RS-S or WS-S, even though the performance decreases with increasing learning complexity, which we quantify by D_{KY} . Similar

to Rathor *et al.*³³ we wanted to rationalize the observations by information processing capacity measures, such as the total information processing capacity. These analyses did not lead to conclusive results for the higher-dimensional tasks.

Thus, we find from the current study that symmetric networks should be chosen for the one-step dynamics prediction in open-loops of these nonlinear dynamical systems with larger numbers of degrees of freedom and $N_{\text{in}} < N_{\text{out}}$. This conclusion might need further exploration in other higher-dimensional systems. However, a similar finding that structured ESN outperforms the RC model with random networks to reconstruct chaotic dynamics in closed loops is reported in Ref. 26 for three-dimensional systems.

The observed trend (based on our discussed learning tasks) that the network becomes increasingly insensitive to its topology as learning complexity grows, raises new questions for spatiotemporal pattern recognition tasks. Perhaps, the current hard-wired, static RC random network architecture is overly simplistic. Incorporating plasticity, which is achieved by an adaptation of the weights of the reservoir network with time, into the RC would likely be necessary to further improve DPT performance.⁶²

ACKNOWLEDGMENTS

The work of S.K.R. and J.S. is funded by the European Union (ERC, MesoComp, No. 101052786). Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. M.Z. acknowledges financial support by Deutsche Forschungsgemeinschaft (DFG, German Research Foundation)—Project RECOMMEND Project No. 536063366 as part of the DFG priority program DFG-SPP 2262 MemrisTec—Project No. 422738993. L.J. is funded by the Carl-Zeiss-Stiftung.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Shailendra K. Rathor: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Lina Jaurigue:** Conceptualization (equal); Formal analysis (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Martin Ziegler:** Conceptualization (equal); Formal analysis (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jörg Schumacher:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Zenodo with record reference No. 17737099, Ref. 65.

APPENDIX A: FURTHER RESULTS FOR THE SF CASE

In the following [Appendices A–D](#), we present the detailed equation systems and more results of the DPTs for the two dynamical systems with the highest number of degrees of freedom, SF and L8. While with increasing number of degrees of freedom the RC will not be able anymore to predict the time series for long times, it can reproduce statistical correlations, as we will demonstrate in the following. To this end, we reconstruct from the time series $A_1(t)$ to $B_4(t)$ in L8 and $a_1(t)$ to $a_9(t)$ in SF the time-dependent velocity and temperature fields to analyse their statistics similar to what was done in Refs. 12, 14, and 63. These results are of primarily fluid mechanical interest. Therefore, they have been moved into the appendices. Furthermore, we underline the importance of a bias b and test if a nonlinear readout improves the performance of the DPT.

1. Shear flow equations

The system of coupled nonlinear ordinary differential equations for the coefficients $a_1(t)$ to $a_9(t)$ in Eq. (11) is given by

$$\frac{da_1}{dt} = \frac{\beta^2}{\text{Re}}(1 - a_1) - \sqrt{\frac{3}{2}} \frac{\beta\gamma}{K_{\alpha\beta\gamma}} a_6 a_8 + \sqrt{\frac{3}{2}} \frac{\beta\gamma}{K_{\beta\gamma}} a_2 a_3, \quad (\text{A1a})$$

$$\frac{da_2}{dt} = -\left(\frac{4\beta^2}{3} + \gamma^2\right) \frac{a_2}{\text{Re}} + \frac{\gamma^2}{\sqrt{6}K_{\alpha\gamma}} \left(\frac{10}{3} a_4 a_6 - a_5 a_7\right) - \frac{\alpha\beta\gamma}{\sqrt{6}K_{\alpha\gamma}K_{\alpha\beta\gamma}} a_5 a_8 - \sqrt{\frac{3}{2}} \frac{\beta\gamma}{K_{\beta\gamma}} (a_1 a_3 + a_3 a_9), \quad (\text{A1b})$$

$$\frac{da_3}{dt} = -\frac{\beta^2 + \gamma^2}{\text{Re}} a_3 + \frac{2}{\sqrt{6}} \frac{\alpha\beta\gamma}{K_{\alpha\gamma}K_{\beta\gamma}} (a_4 a_7 + a_5 a_6) + \frac{\beta^2(3\alpha^2 + \gamma^2) - 3\gamma^2(\alpha^2 + \gamma^2)}{\sqrt{6}K_{\alpha\gamma}K_{\beta\gamma}K_{\alpha\beta\gamma}} a_4 a_8, \quad (\text{A1c})$$

$$\frac{da_4}{dt} = -\frac{3\alpha^2 + 4\beta^2}{3\text{Re}} a_4 - \frac{\alpha}{\sqrt{6}} a_1 a_5 - \frac{10}{3\sqrt{6}} \frac{\alpha^2}{K_{\alpha\gamma}} a_2 a_6 - \sqrt{\frac{3}{2}} \frac{\alpha\beta\gamma}{K_{\alpha\gamma}K_{\beta\gamma}} a_3 a_7 - \sqrt{\frac{3}{2}} \frac{\alpha^2\beta^2}{K_{\alpha\gamma}K_{\beta\gamma}K_{\alpha\beta\gamma}} a_3 a_8 - \frac{\alpha}{\sqrt{6}} a_5 a_9, \quad (\text{A1d})$$

$$\frac{da_5}{dt} = -\frac{\alpha^2 + \beta^2}{\text{Re}} a_5 + \frac{\alpha}{\sqrt{6}} a_1 a_4 + \frac{\alpha^2}{\sqrt{6} \kappa_{\alpha\gamma}} a_2 a_7 - \frac{\alpha\beta\gamma}{\sqrt{6} \kappa_{\alpha\gamma} \kappa_{\alpha\beta\gamma}} a_2 a_8 + \frac{\alpha}{\sqrt{6}} a_4 a_9 + \frac{2}{\sqrt{6}} \frac{\alpha\beta\gamma}{\kappa_{\alpha\gamma} \kappa_{\beta\gamma}} a_3 a_6, \quad (\text{A1e})$$

$$\frac{da_6}{dt} = -\frac{3\kappa_{\alpha\beta\gamma}^2 + \beta^2}{3\text{Re}} a_6 + \frac{\alpha}{\sqrt{6}} a_1 a_7 \sqrt{\frac{3}{2}} \frac{\beta\gamma}{\kappa_{\alpha\beta\gamma}} a_1 a_8 + \frac{10}{3\sqrt{6}} \frac{\alpha^2 - \gamma^2}{\kappa_{\alpha\gamma}} a_2 a_4 - 2\sqrt{\frac{2}{3}} \frac{\alpha\beta\gamma}{\kappa_{\alpha\gamma} \kappa_{\beta\gamma}} a_3 a_5 + \frac{\alpha}{\sqrt{6}} a_7 a_9 + \sqrt{\frac{3}{2}} \frac{\beta\gamma}{\kappa_{\alpha\beta\gamma}} a_8 a_9, \quad (\text{A1f})$$

$$\frac{da_7}{dt} = -\frac{\kappa_{\alpha\beta\gamma}^2}{\text{Re}} a_7 - \frac{\alpha}{\sqrt{6}} (a_1 a_6 + a_6 a_9) + \frac{1}{\sqrt{6}} \frac{\gamma^2 - \alpha^2}{\kappa_{\alpha\gamma}} a_2 a_5 + \frac{1}{\sqrt{6}} \frac{\alpha\beta\gamma}{\kappa_{\alpha\gamma} \kappa_{\beta\gamma}} a_3 a_4, \quad (\text{A1g})$$

$$\frac{da_8}{dt} = -\frac{\kappa_{\alpha\beta\gamma}^2}{\text{Re}} a_8 + \frac{2}{\sqrt{6}} \frac{\alpha\beta\gamma}{\kappa_{\alpha\gamma} \kappa_{\alpha\beta\gamma}} a_2 a_5 + \frac{\gamma^2 (3\kappa_{\alpha\gamma}^2 - \beta^2)}{\sqrt{6} \kappa_{\alpha\gamma} \kappa_{\beta\gamma} \kappa_{\alpha\beta\gamma}} a_3 a_4, \quad (\text{A1h})$$

$$\frac{da_9}{dt} = -\frac{9\beta^2}{\text{Re}} a_9 + \sqrt{\frac{3}{2}} \frac{\beta\gamma}{\kappa_{\beta\gamma}} a_2 a_3 - \sqrt{\frac{3}{2}} \frac{\beta\gamma}{\kappa_{\alpha\beta\gamma}} a_6 a_8, \quad (\text{A1i})$$

where $\kappa_{\alpha\gamma} = \sqrt{\alpha^2 + \gamma^2}$, $\kappa_{\beta\gamma} = \sqrt{\beta^2 + \gamma^2}$, and $\kappa_{\alpha\beta\gamma} = \sqrt{\alpha^2 + \beta^2 + \gamma^2}$. Also $\alpha = 2\pi/L_x$, $\beta = \pi/2$, and $\gamma = 2\pi/L_z$. The latter should not be mixed with the Tikhonov regularization parameter.

2. Statistics of velocity correlations

Figure 11 displays the performance of the R-A network for this three-dimensional shear flow model. It can be observed, how the prediction starts to deviate from the ground truth for the degrees of freedom that have to be reconstructed, caused by the sensitivity to small deviations by round-off errors. This example already

demonstrates that we cannot expect to predict a specific trajectory, as being the case in the L63 system.¹⁸

We probe the accuracy of the prediction of the statistical properties of SF, such as mean basic velocity profile, $\bar{v}_x(y)$, in the wall-normal direction. Such quantities are in view of applications more important than individual system trajectories. Time-dependent flow statistics can be quantified by the following normalized relative error (NARE),^{11,63} which is defined as

$$E[q(y)] = \frac{1}{2\max_{y \in [-1,1]} |q^{GT}(y) - q^P(y)|} \int_{-1}^1 |q^{GT}(y) - q^P(y)| dy. \quad (\text{A2})$$

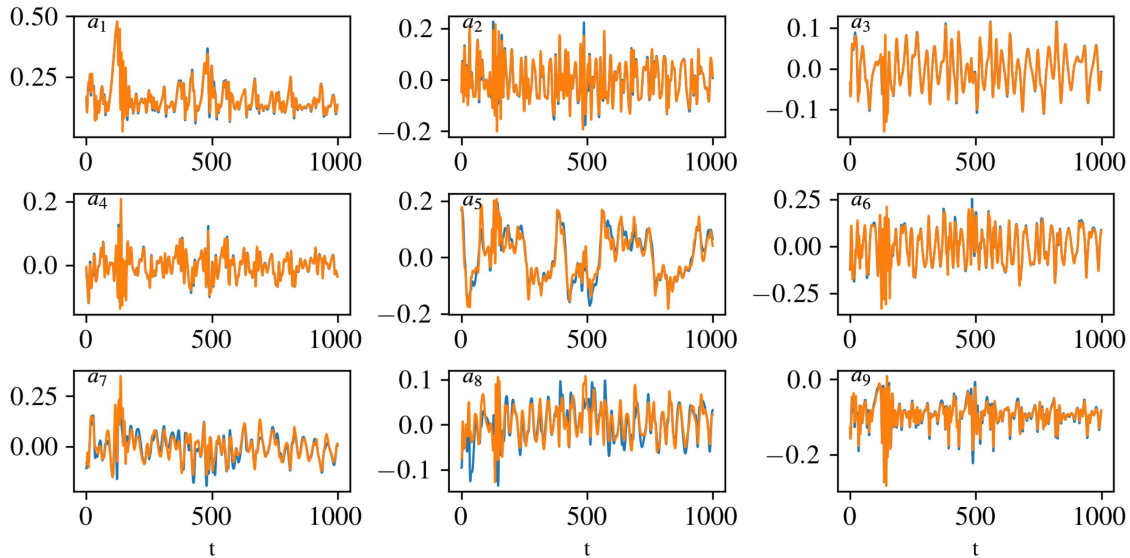


FIG. 11. Predicted time series, $a_1(t)$ to $a_9(t)$ of the three-dimensional plane shear flow (SF) model in Eq. (A1) with a reservoir computing model utilizing the R-A network topology. Solid blue line represents the ground truth, and solid orange line shows the prediction. The number of input modes is five in the open-loop prediction mode. These modes are a_1 , a_2 , a_3 , a_4 , and a_6 . Reconstruction by the reservoir is for all nine degrees of freedom.

TABLE IV. Median of normalized relative error (NARE), as an additional measure of performance next to the MSE. The error is calculated as the median absolute deviation (MAD) of predictions from 1000 differently realized reservoir matrices for a given topology. The quantity is defined by Eq. (A2) for the Galerkin model of a three-dimensional shear flow (SF), which is run in an open loop scenario. Five different quantities are investigated, the streamwise mean flow and 4 important components of the Reynolds stress tensor. See also the corresponding mean vertical profiles, which are displayed in Fig. 12.

	$E[\bar{v}_x(y)]$	$E[\overline{v'_x v'_y}(y)]$	$E[\overline{v'_x v'_x}(y)]$	$E[\overline{v'_y v'_y}(y)]$	$E[\overline{v'_z v'_z}(y)]$
R-A	0.05 ± 0.03	0.5 ± 0.3	3.0 ± 0.3	5.1 ± 0.7	4.5 ± 1.4
RS-A	0.08 ± 0.05	0.9 ± 0.5	2.0 ± 0.6	1.1 ± 0.7	1.6 ± 0.9
RS-S	0.06 ± 0.03	0.4 ± 0.3	2.4 ± 0.2	3.5 ± 0.7	1.6 ± 0.7
WS-A	0.07 ± 0.04	1.1 ± 0.6	2.0 ± 0.5	1.2 ± 0.6	2.0 ± 0.9
WS-S	0.06 ± 0.03	0.4 ± 0.3	2.4 ± 0.2	3.5 ± 0.7	1.6 ± 0.7

Here, q is the quantity of interest, and q^{GT} and q^p are the ground truth and prediction. Table IV shows that the outcome with respect to NARE is slightly different to that for the MSE. With respect to a reconstruction of the shear flow in a statistical sense, the R-A network performs best, with $E[\bar{v}_x(y)] = 0.05\%$.

Figure 12 displays some important statistical correlations of SF. The vertical (wall-normal) mean flow profile in panel (a) is well reconstructed for all five cases. In panels (b)–(e) of the same figure, we provide the vertical mean profiles of four components of the Reynolds stress tensor $\overline{v'_i v'_j}$ in Eq. (A3), which quantifies the correlations between the velocity components. It is seen that first- and second-order moments of the velocity fields are reproduced fairly

well by the RC for all five topologies. The strongest scatter arises for the autocorrelations of the spanwise velocity component with the prediction by R-A topology closest to the ground truth. The components of the Reynolds stress tensor are given by

$$\overline{v'_i v'_j}(y) = \frac{1}{L_x L_z} \int_0^{L_z} \int_0^{L_x} \langle v'_i v'_j \rangle_t dx dz. \quad (A3)$$

They are obtained by the Reynolds decomposition of the velocity field components

$$v'_i(x, y, z, t) = v_i(x, y, z, t) - \langle v_i(x, y, z) \rangle_t, \quad (A4)$$

where $\langle \cdot \rangle_t$ denotes averaging with respect to time.

APPENDIX B: FURTHER RESULTS FOR THE L8 CASE

1. Extended Lorenz model equations

The system of nonlinear ordinary differential equations for the coefficients $A_1(t)$ to $B_4(t)$ in Eq. (8) is given by

$$\frac{dA_1}{d\tau} = \sigma(B_1 - A_1) - \frac{c_3}{\sqrt{2}c_1} A_2 A_3 + \frac{3c_{-5}}{\sqrt{2}c_1} A_3 A_4, \quad (B1a)$$

$$\frac{dB_1}{d\tau} = -B_1 + rA_1 - A_1 B_2 + \frac{1}{2} A_2 B_3 + \frac{3}{2} A_4 B_3, \quad (B1b)$$

$$\frac{dB_2}{d\tau} = -bB_2 + A_1 B_1, \quad (B1c)$$

$$\frac{dA_2}{d\tau} = -\frac{\sigma b}{4} A_2 - \frac{3}{2\sqrt{2}} A_1 A_3, \quad (B1d)$$

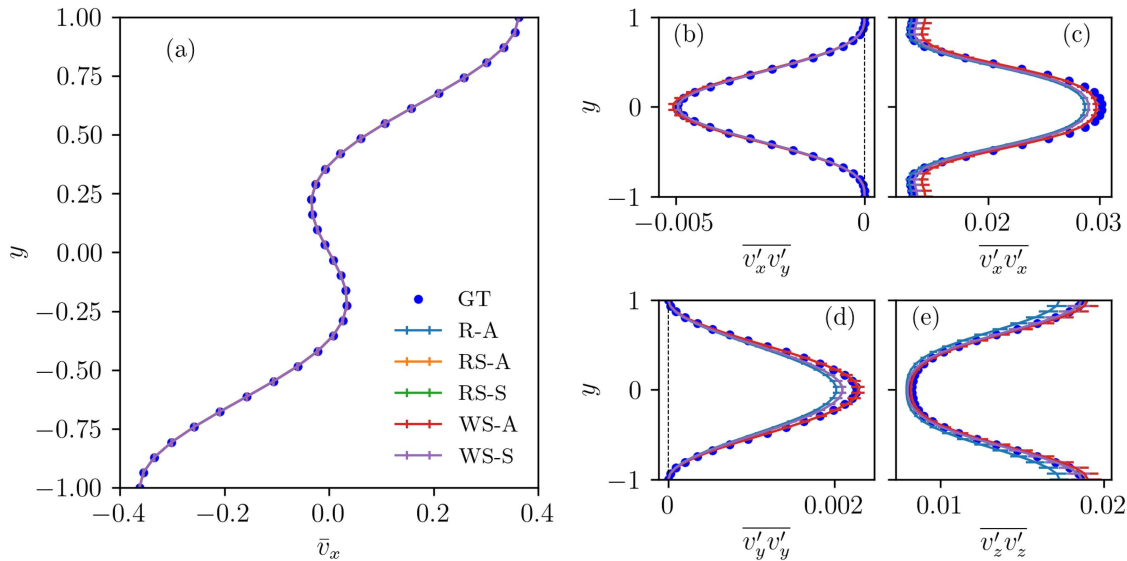


FIG. 12. Fluid flow statistics of the three-dimensional shear flow model. (a) Comparison of the predicted mean streamwise velocity component profile obtained from different networks with the ground truth (GT). (b)–(e) Comparison of predicted mean Reynolds stress profiles for different networks with the GT. All profiles are averages with respect to time as well as streamwise and spanwise directions. The error bars are the standard deviations over predictions from 1000 differently realized reservoir matrices for a given topology. Note that the Reynolds stress profiles $\overline{v'_y v'_y}$ and $\overline{v'_x v'_x}$ have to be zero at $y = \pm 1$ due to the free-slip boundary conditions for the velocity field.

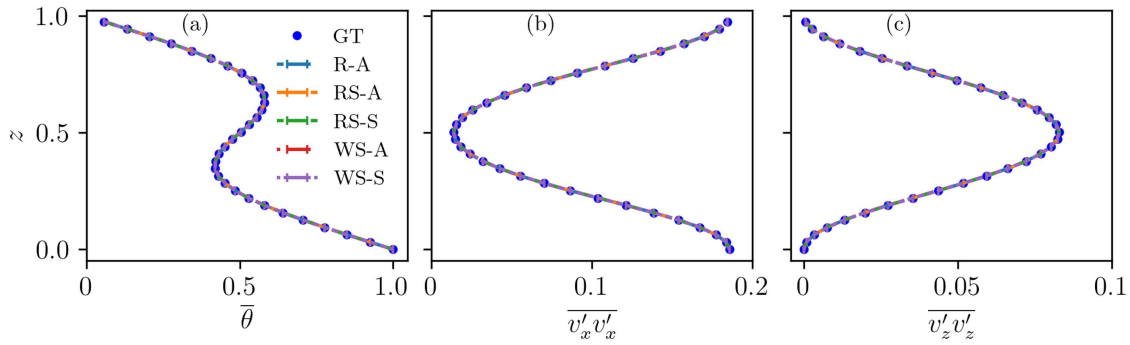


FIG. 13. Statistics of the extended Lorenz-type model in Eq. (B1). (a) Comparison of predicted mean thermal fluctuations obtained from different networks with the ground truth (GT). (b) and (c) Comparison of predicted mean Reynolds stress profiles for different networks with the GT. All profiles are averages with respect to time as well as the x -direction. The error bars are the standard deviations over predictions from 1000 differently realized reservoir matrices for a given topology. The Reynolds stress profiles $\overline{v'_z v'_z}$ have to be zero at $z = 0, 1$ due to the free-slip boundary conditions.

$$\frac{dA_3}{d\tau} = -\sigma \left(\frac{c_4}{c_1} A_3 + \frac{c_1}{\sqrt{2}(c_4)} B_3 \right) + \frac{c_0}{\sqrt{2}c_1} A_1 A_2 - \frac{3c_{-8}}{\sqrt{2}(c_4)} A_1 A_4, \quad (\text{B1e})$$

$$\frac{dA_4}{d\tau} = -\frac{9\sigma b}{4} A_4 - \frac{1}{2\sqrt{2}} A_1 A_3, \quad (\text{B1f})$$

$$\frac{dB_3}{d\tau} = -\frac{c_4}{c_1} B_3 - A_2 B_3 + \sqrt{2} r A_3 + 3A_4 B_1 - 2\sqrt{2} A_3 B_4, \quad (\text{B1g})$$

$$\frac{dB_4}{d\tau} = -4bB_4 + \frac{3\sqrt{2}}{4} A_3 B_3, \quad (\text{B1h})$$

where $c_n = \alpha^2 + n\beta^2$ ($n \in \mathbb{Z}$), Prandtl number $\sigma = 10$, $b = 8/3$, and relative Rayleigh number $r = 28$. The rescaled time is $\tau = c_1 t$.

2. Statistics of velocity and temperature correlations

For completeness, we repeated this analysis for the L8 model. Figure 13 reports some results for the L8 model. We show the mean temperature profile taken across the convection layer together with the profiles of the velocity fluctuations in x and z directions, the diagonal entries of the Reynolds stress tensor. A very good agreement of all five networks with the ground truth can be observed. The number of nodes in all reservoir networks was $N = 1024$. The standard deviation error from the predictions of 1000 differently initialized reservoir coupling matrices is $O(10^{-6})$. We see in both cases that the RC models can be used as surrogate models for the fluid flows.

APPENDIX C: ABSENCE OF BIAS IN A RESERVOIR COMPUTING SETUP

The input bias vector $\mathbf{b}$ in Eq. (1) is realized by populating the elements either from a uniform random distribution in $[-0.5, 0.5]$ or with constant value. In Fig. 14, the variation of MSE with respect to network topology for the systems L8 and SF is shown. Here, grid search is carried out to tune hyperparameters ($\rho, \varepsilon, \gamma$) for each task with the same Δt as in Subsection IV A, and reservoir size $N = 1024$

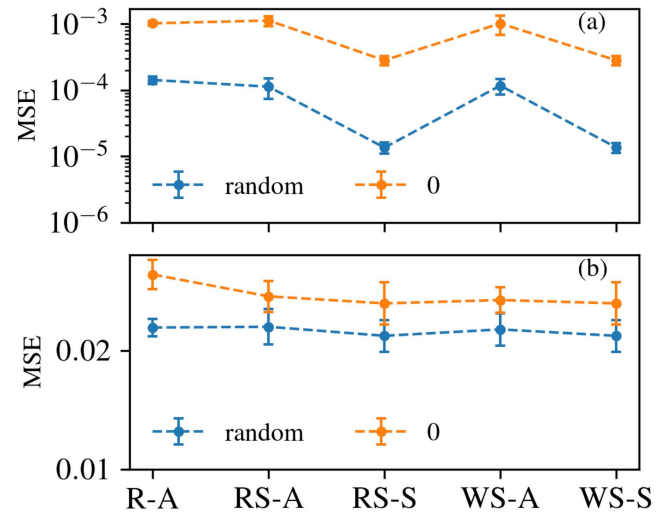


FIG. 14. Effect of input bias on the performance of (a) L8 and (b) SF systems. The legend shows the realization of the input bias: either random or $\mathbf{b} = 0$ (no bias).

and density $D_r = 0.008$ are fixed. We find the best performance for both L8 and SF systems for the input bias with random elements from a uniform random distribution.

APPENDIX D: NONLINEAR READOUT SCHEME

The readout scheme in our study is given in Eq. (2) and can be read as

$$\mathbf{y}^p(k) = W^{\text{out}}[r_1(k), r_2(k), r_3(k), r_4(k), \dots]^T. \quad (\text{D1})$$

This is a *linear* readout scheme. As an alternative approach to break symmetry that may lead to creation of mirror-attractor in the RC space, a nonlinear readout with quadratic terms is proposed.^{36,37,64} We test this approach for the L8 model. To this end, the readout scheme is designed by taking square of the alternate element of

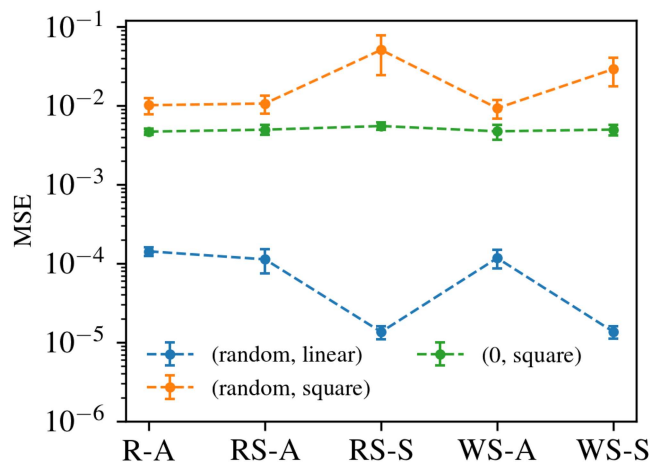


FIG. 15. Effect of the readout scheme on the performance of L8. The legend shows the pair of input bias and readout scheme.

reservoir state vector as follows:

$$\mathbf{y}^p(k) = \mathbf{W}^{\text{out}} [r_1(k), r_2^2(k), r_3(k), r_4^2(k), \dots]^T. \quad (\text{D2})$$

This is a *square* readout scheme. This choice is such that the number of parameters in $\mathbf{W}^{\text{out}}$ remains the same for both *linear* and *square* readout schemes. Here, with fixed reservoir size $N = 1024$ and density $D_r = 0.008$, grid search is performed to tune hyperparameters ($\rho, \varepsilon, \gamma$) for each task with $\Delta t = 0.1$ as in Subsection IV A. In Fig. 15, considering these schemes coupled with input bias, we find that linear readout with random input bias produces best results for all network types.

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